PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

IDENTIFICATION OF URL BASED ATTACKS FROM IP DATA

A PROJECT REPORT

Submitted by

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IN

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this project titled "IDENTIFICATION OF URL BASED ATTACKS FROM IP DATA" is a bonafide work of "K.MANOHITH (20221CSE0199), V.SIVANANDAREDDY (20221CSE0245), and P.SREEDHAR RAO (20221CSE0221)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING during 2025-26.

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DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named K. Manohith, V. Sivananda Reddy , P. Sreedhar Rao, hereby declare that the project work titled "**IDENTIFICATION OF URL BASED ATTACKS FROM IP DATA**" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING (Spl. Division) during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

This project focuses on identifying malicious URLs using a modern deep learning approach. With the rapid increase in internet usage, cyber threats such as phishing, malware, spam, and website defacement have become more common, often targeting users through deceptive web links. Detecting such harmful URLs in real time is a critical challenge in cybersecurity.

To address this issue, the proposed system uses a hybrid deep learning model that combines Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks. This combination allows the system to capture both hidden patterns within URLs and the sequence-based relationships in their structure.Along with analyzing the URL itself, the model also considers IP-related information to improve detection accuracy.

The system processes URL data through preprocessing, feature extraction, and tokenization before feeding it into the model for classification. It then categorizes URLs into different types such as benign, phishing, malware, spam, and defacement. A simple web-based interface is also developed to allow users to test URLs in real time.

Overall, the proposed approach provides a more intelligent and reliable way to detect malicious URLs compared to traditional methods. It helps improve online security by reducing the risk of cyber-attacks and supporting safer internet usage.